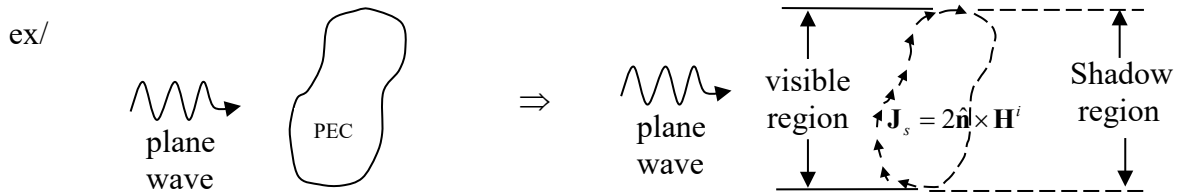


Notes 9

Physical Optics – an approximation for the current when the equivalence principle is used in scattering situations

$$\hat{\mathbf{J}}_s = 2\hat{\mathbf{n}} \times \mathbf{H}^i \text{ in the "visible" region}$$

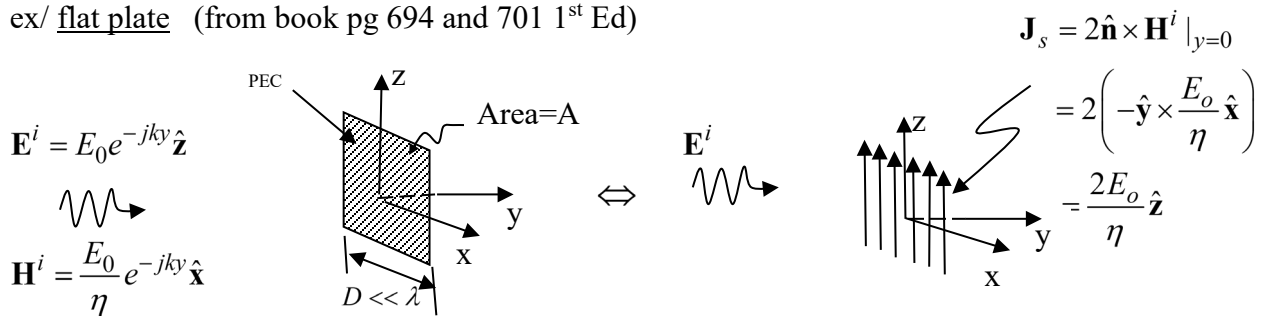


This current is inserted into the free space potential equation to compute the scattered field

$$\mathbf{A} = \frac{\mu}{4\pi} \int_{s'} \left(2\hat{\mathbf{n}} \times \mathbf{H}^i \right) \frac{e^{-jkR}}{R} ds' \quad \rightarrow \quad \mathbf{E}^{scat} = \frac{-j\omega}{k^2} \left(k^2 \mathbf{A} + \nabla \nabla \cdot \mathbf{A} \right)$$

Note: Physical Optics works best if the object is very large in terms of wavelengths, maximum diameter D at least $> 2\lambda$. However all the examples that follow, found in Harrington, A.L. Maffett, and Bowman, Senior and Uslenghi, are for $D \ll \lambda$, so the answer is considered to be a '1st order approximation'.

ex/ flat plate (from book pg 694 and 701 1st Ed)



$$\begin{aligned} \mathbf{E}^s &= j\omega A_z \sin \theta \hat{\boldsymbol{\theta}} \\ &= j\omega\mu \frac{1}{4\pi} \int_{s'} \left(\frac{2E_0}{\eta} \right) \frac{e^{-jkr}}{r} dx' dz' \sin \theta \hat{\boldsymbol{\theta}} \quad \text{because } D \ll \lambda \text{ the next term is ignored as was done with the short dipole} \\ &= j \underbrace{\frac{\omega\mu}{\eta}}_{=k} \frac{E_0}{2\pi} \frac{e^{-jkr}}{r} \sin \theta \underbrace{\int_{s'} dx' dz'}_{=A} \hat{\boldsymbol{\theta}} \\ &= \underline{\underline{\frac{jkE_0 A}{2\pi} \frac{e^{-jkr}}{r} \sin \theta \hat{\boldsymbol{\theta}}}} \end{aligned}$$

(RCS) Radar Cross Section (Echo Area)

$$A_e = \lim_{r \rightarrow \infty} \left(4\pi r^2 \frac{S^s}{S^i} \right), \quad S = \left| \frac{1}{2} \text{Re} \{ \mathbf{E} \times \mathbf{H}^* \} \right|$$

Where $\mathbf{S}$ is the power density of the field traveling back in the direction of the transmitter.

ex/ Backscatter RCS of a flat plate (with Physical Optics approximation).

Backscatter direction $\Rightarrow \theta = 90^\circ, \phi = 270^\circ$

$$\mathbf{E}^s = \frac{jkE_0 A}{2\pi r} \underbrace{e^{-jkr}}_{=1} \underbrace{\sin \theta}_{\substack{\text{when} \\ \theta=90^\circ}} \hat{\boldsymbol{\theta}} = -\hat{\mathbf{z}}$$

$$S^i = \frac{|E_0|^2}{2\eta}$$

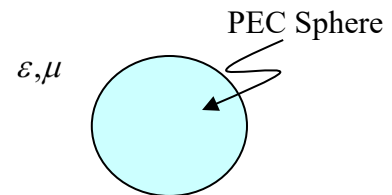
$$S^s = \frac{1}{2\eta} \left| \frac{kE_0 A}{2\pi r} \right|^2$$

$$A_e = \lim_{r \rightarrow \infty} \left(4\pi r^2 \frac{\frac{1}{2\eta} \left| \frac{kE_0 A}{2\pi r} \right|^2}{\frac{|E_0|^2}{2\eta}} \right) = \frac{4\pi A^2}{\lambda^2}$$

ex/ Small (radius $\ll \lambda$) PEC Sphere (Maffett – Topics for a Statistical Description of RCS Wiley)

backscatter RCS:

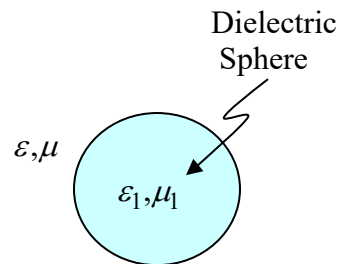
$$A_e = \frac{9\lambda^2}{4\pi} \left(\frac{2\pi a}{\lambda} \right)^6, \quad a \equiv \text{radius of the sphere}$$



ex/ Small (radius $\ll \lambda$) dielectric sphere – (Shen & Kong, Applied Electromagnetism pg. 215)

backscatter RCS:

$$\begin{aligned} A_e &= 4\pi \left(\frac{\epsilon_1 - \epsilon}{\epsilon_1 + 2\epsilon} \right)^2 k^4 a^6 \\ &= \frac{\lambda^2}{\pi} \left(\frac{\epsilon_1 - \epsilon}{\epsilon_1 + 2\epsilon} \right)^2 \left(\frac{2\pi a}{\lambda} \right)^6 \end{aligned}$$



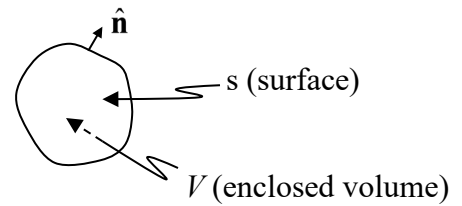
Green's Theorem – a method for changing a differential equation to an integral equation.

- Divergence theorem:

$$\int_V \nabla \cdot \mathbf{u}(\mathbf{r}) dv = \oint_s \mathbf{u}(\mathbf{r}) \cdot \hat{\mathbf{n}} ds \quad (1)$$

- let $\mathbf{u} = \psi \nabla \phi$

so
$$\begin{aligned} \nabla \cdot \mathbf{u} &= \nabla \cdot (\psi \nabla \phi) = \nabla \psi \cdot \nabla \phi + \psi \nabla \cdot \nabla \phi \\ &= \nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi \end{aligned} \quad (2)$$



Note that

$$\nabla \phi \cdot \hat{\mathbf{n}} = \frac{\partial \phi}{\partial n} \quad (3)$$

is the derivative in the direction of the unit vector $\hat{\mathbf{n}}$ which in this case is normal to the surface.

Substitute (2) and (3) into (1)

$$\int_V (\nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi) dv = \oint_s \psi \frac{\partial \phi}{\partial n} ds \quad (4)$$

Green's 1st identity

Interchange ψ and ϕ in (4) to get

$$\int_V (\nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi) dv = \oint_s \phi \frac{\partial \psi}{\partial n} ds \quad (5)$$

Subtract (5) from (4) to get

$$\int_V (\psi \nabla^2 \phi - \phi \nabla^2 \psi) dv = \oint_s \left(\psi \frac{\partial \phi}{\partial n} - \phi \frac{\partial \psi}{\partial n} \right) ds \quad (6)$$

Green's 2nd identity (or Green's theorem)

- Green's function solution to Helmholtz magnetic vector potential equation:

$$\nabla'^2 A(\mathbf{r}') + k^2 A(\mathbf{r}') = -\mu J(\mathbf{r}') \quad (7)$$

$$\nabla'^2 G(\mathbf{r}, \mathbf{r}') + k^2 G(\mathbf{r}, \mathbf{r}') = -\delta(\mathbf{r} - \mathbf{r}') \quad (8)$$

r observation variable

r' source variable

Note: A and J are functions of r'

G is a function of r and r'

Multiply $G \times (7)$ and $A \times (8)$ and subtract

$$G \nabla'^2 A - A \nabla'^2 G = -\mu JG + A \delta(\mathbf{r} - \mathbf{r}')$$

Notice that the derivatives are over primed coordinates !! Integrate over a volume of space v'

$$\int_{v'} (G \nabla'^2 A - A \nabla'^2 G) dv' = \mu \int_{v'} (-JG) dv' + \int_{v'} A \delta(\mathbf{r} - \mathbf{r}') dv' \quad (9)$$

Use Green's
2nd identity

$$\oint_{s'} \left(G \frac{\partial A}{\partial n'} - A \frac{\partial G}{\partial n'} \right) ds' = -\mu \int_{v'} JG dv' + \int_{v'} A \delta(\mathbf{r} - \mathbf{r}') dv' \quad (10)$$

If A is known on $s' \rightarrow$ Dirichlet boundary condition

If $\frac{\partial A}{\partial n'}$ is known on $s' \rightarrow$ Neumann boundary condition

If A and $\frac{\partial A}{\partial n'}$ are known on $s' \rightarrow$ Cauchy boundary condition

Let $s' \rightarrow \infty \Rightarrow$ 1st integral $\rightarrow 0$ because A , and therefore $G \rightarrow 0$ at ∞ due to Sommerfeld radiation condition.

$$0 = -\mu \int_{v'} J(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv' + \int_{v'} A(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}') dv'$$

$$A(\mathbf{r}) = \mu \int_{v'} J(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv' \quad (11)$$

Interpretation :
sum over weighted
sources

Note: Although v' includes all of space, the integral is evaluated only over the source ($J(\mathbf{r}')$) distribution region because $J(\mathbf{r}')$ is zero outside this region.

Sommerfeld Radiation Condition – for Helmholtz equation in n-dimensional space.

$$\lim_{R \rightarrow \infty} R^{\frac{n-1}{2}} \left(\frac{\partial \psi}{\partial R} + jk\psi \right) = 0$$

$n=1, 2, \text{ or } 3 \text{ dimensions}$

where

$R = x, y, \text{ or } z \text{ when } n = 1$

$R = \rho \text{ when } n = 2$

$R = r \text{ when } n = 3$

See Stratton pg 486, Stensen or another book by Papas. See proof in Goodman, Fourier Optics pg 99.

Also see other conditions in Stratton (Regular at infinity, etc)